

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Botany

Paper: BOB-603 — Cytogenetics and Evolutionary Processes

Time: Three hours

Maximum Marks: 70

Instructions: Answer five questions in all, selecting two each from Sections A and B and Section-C which is compulsory. The figures in the right-hand margin indicate marks.

SECTION A

Question 1. Describe the various types of chromosomal aberrations with suitable examples.
[15]

Question 2. Describe any two of the following: [2 × 7.5 = 15]

- (a) Lampbrush chromosome
- (b) Chemical composition of chromosome
- (c) Difference between inversion and translocation with examples

Question 3. What is karyotype? How can it be used to describe evolution? Illustrate with suitable examples. [15]

Question 4. Write critical notes on any three of the following: [3 × 5 = 15]

- (a) Linkage map
- (b) Chromatin and its types
- (c) Heteroploidy
- (d) Different types of trisomic

SECTION B

Question 5. Explain how recombination frequency can be used to determine gene positions on a chromosome. Illustrate with examples. [15]

Question 6. Explain any two of the following: [2 × 7.5 = 15]

- (a) Deletion mapping
- (b) Somatic cell fusion and selection of hybrid cells
- (c) Difference between alkylating and intercalating agents

Question 7. What is cytoplasmic inheritance? Describe cytoplasmic inheritance with suitable examples. [15]

Question 8. Discuss any two of the following: $[2 \times 7.5 = 15]$

- (a) Difference between qualitative and quantitative trait inheritance
- (b) Multiple allelism
- (c) Mitochondrial and chloroplast DNA

SECTION C

Question 9. Write short notes on the following in not more than 50 words each: $[10 \times 1 = 10]$

- (a) Base analogs
- (b) 6,4-PP
- (c) Radiation hybrids
- (d) GISH
- (e) Reciprocal cross
- (f) *Brassicoraphanus*
- (g) Centromere
- (h) Eukaryotic organelles involved in cytoplasmic inheritance
- (i) Biochemical nature of Rh factor
- (j) Self-incompatibility.

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